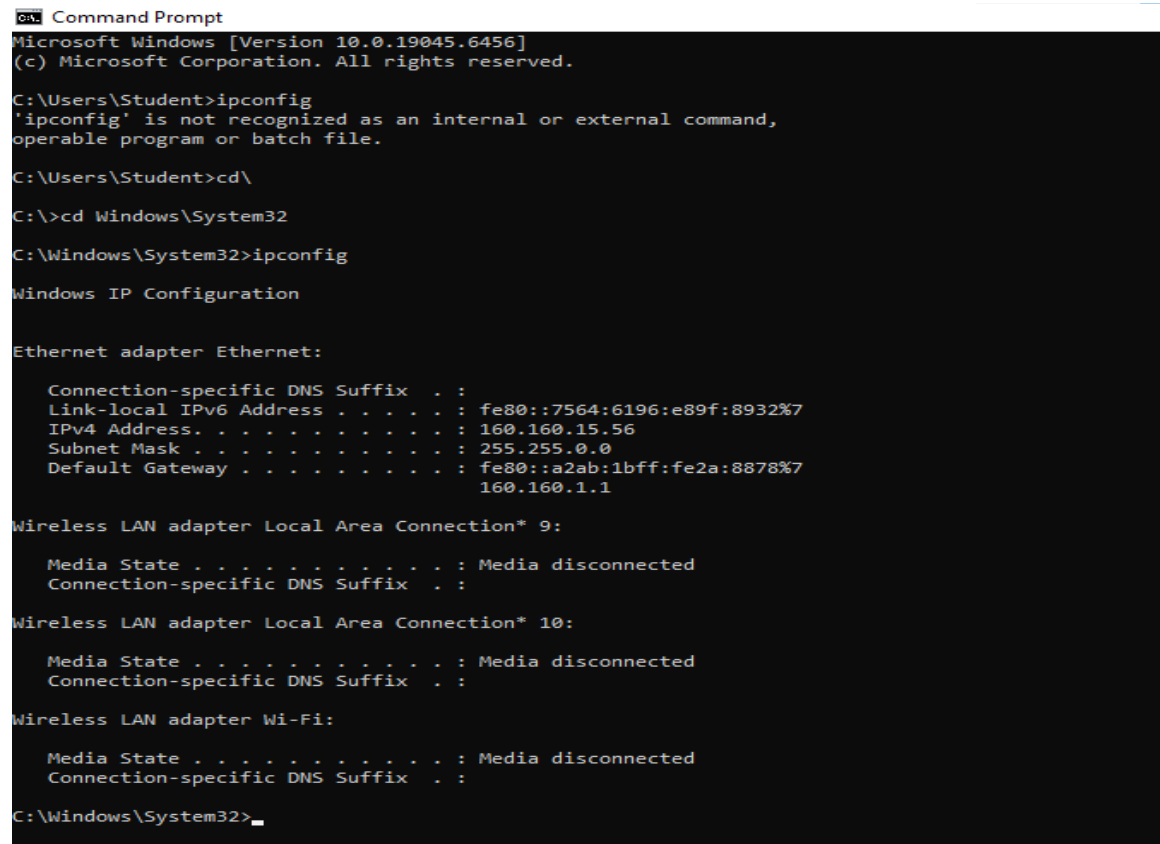


Practical 3 : To study of various networking commands.

Retrieving Information about Your Network Connection

1. ipconfig : The **ipconfig** command is used in **Command Prompt** to display network information such as the **IP address, default gateway, subnet mask, and network adapter status** of a computer. It helps users check and troubleshoot network connections.



```

C:\> Command Prompt
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Student>ipconfig
'ipconfig' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\Student>cd\

C:\>cd Windows\System32

C:\Windows\System32>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::7564:6196:e89f:8932%7
    IPv4 Address. . . . . : 160.160.15.56
    Subnet Mask . . . . . : 255.255.0.0
    Default Gateway . . . . . : fe80::a2ab:1bff:fe2a:8878%7
                                160.160.1.1

Wireless LAN adapter Local Area Connection* 9:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 10:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

C:\Windows\System32>_
  
```

2. ipconfig /renew : The **ipconfig /renew** command is used to request and obtain a new IP address from the network for the computer.

3. ipconfig /release : The **ipconfig /release** command is used to release the current IP address assigned to the computer and disconnect it from the network temporarily.

4. ipconfig /all : The **ipconfig /all** command is used to display complete network configuration details, including the **IP address, MAC address, DNS server, subnet mask, and default gateway** of the computer.

Command Prompt

```
C:\Windows\System32>ipconfig /all

Windows IP Configuration

Host Name . . . . . : DESKTOP-CFFK9F2
Primary Dns Suffix . . . . . :
Node Type . . . . . : Mixed
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet:

Connection-specific DNS Suffix . :
Description . . . . . : Realtek PCIe GbE Family Controller
Physical Address. . . . . : 48-4D-7E-BC-F8-A7
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::7564:6196:e89f:8932%7(Preferred)
IPv4 Address. . . . . : 160.160.15.56(Preferred)
Subnet Mask . . . . . : 255.255.0.0
Default Gateway . . . . . : fe80::a2ab:1bff:fe2a:8878%7
                          160.160.1.1
DHCPv6 IAID . . . . . : 340282750
DHCPv6 Client DUID. . . . . : 00-01-00-01-2D-FB-AB-7F-48-4D-7E-BC-F8-A7
DNS Servers . . . . . : 160.160.1.1
NetBIOS over Tcpip. . . . . : Enabled

Wireless LAN adapter Local Area Connection* 9:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . : 1A-0F-99-7E-E2-47
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 10:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
Physical Address. . . . . : 2A-0F-99-7E-E2-47
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Dell Wireless 1707 802.11b|g|n (2.4GHZ)
Physical Address. . . . . : D8-0F-99-7E-E2-47
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
```

Finding Your Computer's MAC Address

5. getmac : The **getmac** command is used to display the **MAC (Media Access Control)** address of the network adapters installed on the computer.

```
C:\Windows\System32>getmac

Physical Address      Transport Name
=====
48-4D-7E-BC-F8-A7    \Device\Tcpip_{713B71ED-BE4D-43A1-9721-73F101DBFD4F}
D8-0F-99-7E-E2-47    Media disconnected
```

Check Network Connections and Display Port Information

6. netstat : The **netstat** command is used to display detailed information about a computer's network activity. It shows **active network connections, open ports, IP addresses, and network statistics**. It also helps users monitor network traffic and troubleshoot connection-related problems. (**netstat /a:** The **netstat /a** command is used to display **all active connections and listening ports** on the computer.)

```
C:\Windows\System32>netstat
```

Active Connections

Proto	Local Address	Foreign Address	State
TCP	160.160.15.56:1521	DESKTOP-CFFK9F2:49675	ESTABLISHED
TCP	160.160.15.56:7680	160.160.12.12:50894	TIME_WAIT
TCP	160.160.15.56:7680	DESKTOP-709BNBT:52159	TIME_WAIT
TCP	160.160.15.56:7680	160.160.107.222:61172	TIME_WAIT
TCP	160.160.15.56:7680	160.160.110.8:56888	TIME_WAIT
TCP	160.160.15.56:7680	160.160.110.220:53615	TIME_WAIT
TCP	160.160.15.56:7680	160.160.110.220:53650	TIME_WAIT
TCP	160.160.15.56:7680	160.160.110.220:53662	TIME_WAIT
TCP	160.160.15.56:7680	160.160.112.240:58758	TIME_WAIT
TCP	160.160.15.56:49675	DESKTOP-CFFK9F2:1521	ESTABLISHED
TCP	160.160.15.56:57957	4.213.25.240:https	ESTABLISHED
TCP	160.160.15.56:58097	se-in-f188:https	ESTABLISHED
TCP	160.160.15.56:58115	WIN-CEIT:microsoft-ds	ESTABLISHED
TCP	160.160.15.56:58156	a23-196-253-238:https	CLOSE_WAIT
TCP	160.160.15.56:58750	pnbomb-bb-in-f3:https	ESTABLISHED
TCP	160.160.15.56:58766	tzdelb-bg-in-f5:https	ESTABLISHED
TCP	160.160.15.56:58769	57.155.141.122:https	TIME_WAIT
TCP	160.160.15.56:58771	4.230.171.124:https	TIME_WAIT

^C

```
C:\Windows\System32>netstat /a
```

Active Connections

Proto	Local Address	Foreign Address	State
TCP	0.0.0.0:135	DESKTOP-CFFK9F2:0	LISTENING
TCP	0.0.0.0:445	DESKTOP-CFFK9F2:0	LISTENING
TCP	0.0.0.0:1521	DESKTOP-CFFK9F2:0	LISTENING
TCP	0.0.0.0:5040	DESKTOP-CFFK9F2:0	LISTENING
TCP	0.0.0.0:7680	DESKTOP-CFFK9F2:0	LISTENING
TCP	0.0.0.0:49664	DESKTOP-CFFK9F2:0	LISTENING
TCP	0.0.0.0:49665	DESKTOP-CFFK9F2:0	LISTENING
TCP	0.0.0.0:49666	DESKTOP-CFFK9F2:0	LISTENING
TCP	0.0.0.0:49667	DESKTOP-CFFK9F2:0	LISTENING
TCP	0.0.0.0:49668	DESKTOP-CFFK9F2:0	LISTENING
TCP	0.0.0.0:49674	DESKTOP-CFFK9F2:0	LISTENING
TCP	0.0.0.0:49676	DESKTOP-CFFK9F2:0	LISTENING
TCP	127.0.0.1:8080	DESKTOP-CFFK9F2:0	LISTENING
TCP	127.0.0.1:49670	DESKTOP-CFFK9F2:0	LISTENING
TCP	160.160.15.56:139	DESKTOP-CFFK9F2:0	LISTENING
TCP	160.160.15.56:1521	DESKTOP-CFFK9F2:49675	ESTABLISHED
TCP	160.160.15.56:7680	DESKTOP-CFFK9F2:53217	TIME_WAIT
TCP	160.160.15.56:7680	DESKTOP-CFFK9F2:53227	TIME_WAIT
TCP	160.160.15.56:7680	DESKTOP-709BNBT:52322	TIME_WAIT
TCP	160.160.15.56:7680	160.160.105.142:56182	TIME_WAIT

^C

7. ping : To test your connection to a certain **website** or another **network** or **Internet location**, you can use the **ping** network command followed by a **web-address** or **IP address**.

ping 160.160.15.254 and ping 160.160.15.58

```
C:\Windows\System32>ping 160.160.15.254
```

```
Pinging 160.160.15.254 with 32 bytes of data:
Reply from 160.160.15.254: bytes=32 time<1ms TTL=128
Reply from 160.160.15.254: bytes=32 time<1ms TTL=128
Reply from 160.160.15.254: bytes=32 time<1ms TTL=128
Reply from 160.160.15.254: bytes=32 time<1ms TTL=128

Ping statistics for 160.160.15.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\Windows\System32>ping 160.160.15.58
```

```
Pinging 160.160.15.58 with 32 bytes of data:
Reply from 160.160.15.58: bytes=32 time<1ms TTL=128
Reply from 160.160.15.58: bytes=32 time<1ms TTL=128
Reply from 160.160.15.58: bytes=32 time<1ms TTL=128
Reply from 160.160.15.58: bytes=32 time<1ms TTL=128

Ping statistics for 160.160.15.58:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

ping www.google.com

```
C:\Windows\System32>ping www.google.com

Pinging www.google.com [142.251.155.119] with 32 bytes of data:
Reply from 142.251.155.119: bytes=32 time=11ms TTL=118
Reply from 142.251.155.119: bytes=32 time=12ms TTL=118
Reply from 142.251.155.119: bytes=32 time=12ms TTL=118
Reply from 142.251.155.119: bytes=32 time=11ms TTL=118

Ping statistics for 142.251.155.119:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 11ms, Maximum = 12ms, Average = 11ms
```

ping ldrp.ac.in

```
C:\Windows\System32>ping ldrp.ac.in

Pinging ldrp.ac.in [178.105.92.197] with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 178.105.92.197:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

8. hostname : The **hostname** command is used to display the **name of the computer (host name)** currently logged into the system.

```
C:\Windows\System32>hostname
DESKTOP-CFFK9F2
```

9. pathping : The **pathping** command is used to find **network latency** and **packet loss** between the computer and a destination. It helps analyze network performance.

```
C:\Windows\System32>pathping google.com

Tracing route to google.com [192.178.174.113]
over a maximum of 30 hops:
  0  DESKTOP-CFFK9F2 [160.160.15.56]
  1  160.160.1.1
  2  10.119.237.17
  3  10.154.7.45
  4  10.255.236.161
  5  10.152.7.214
  6  142.250.172.80
  7  72.14.238.215
  8  142.251.251.186
  9  ^C
```

10. nslookup : Displays information that you can use to diagnose Domain Name System (DNS) infrastructure.

```
C:\Windows\System32>nslookup www.google.com
Server:      UnKnown
Address:     160.160.1.1

Non-authoritative answer:
Name:        www.google.com
Addresses:   2001:4860:4828:7700::
             2001:4860:4827:7700::
             2001:4860:482c:7700::
             2001:4860:482b:7700::
             2001:4860:4826:7700::
             2001:4860:482a:7700::
             2001:4860:4829:7700::
             2001:4860:482d:7700::
             142.251.155.119
             142.251.153.119
             142.251.157.119
             142.251.151.119
             142.251.150.119
             142.251.152.119
             142.251.154.119
             142.251.156.119
```

11. tracert : Show the path a packet of information takes from your computer to the specified host.

```
C:\Windows\System32>tracert www.google.com

Tracing route to www.google.com [142.251.152.119]
over a maximum of 30 hops:

  1  <1 ms    <1 ms    <1 ms    160.160.1.1
  2  <1 ms    <1 ms    <1 ms    10.119.237.17
  3  16 ms    17 ms    16 ms    10.154.7.45
  4  17 ms    16 ms    16 ms    10.255.236.161
  5  13 ms    12 ms    13 ms    10.152.7.214
  6  20 ms    18 ms    16 ms    72.14.204.62
  7  19 ms    18 ms    18 ms    142.251.152.119

Trace complete.

C:\Windows\System32>
```

Lab Exercise :

1. Why are we using *ping* command ?

2. Explain *arp* command?
